# Temperature Stability and Flux Losses Over Time in Sintered Nd–Fe–B Permanent Magnets

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**Demagnetization occurring with time in a constant temperature is often assumed to be negligible, provided that the operating point of the magnet is above the knee of the $B$–$H$ curve. However, because it is important to ensure that no permanent flux losses occur in permanent magnets during their use in industrial applications, there is a clear need to quantify these time-dependent losses. We measured losses over time for four commercial sintered Nd–Fe–B magnet grades at five different temperatures (23 °C–150 °C). We tested samples with three different permeance coefficient (Pc) values for each material. The time-dependent losses measured fitted the logarithmic law of magnetic viscosity well. We demonstrated that the total flux loss in a lifetime of 30 years can be estimated according to the temperature, coercivity of the material, and the permeance coefficient of the magnet. With proper selection of the magnet material, in accordance with the designed Pc of the application, the total flux loss in 30 years can be minimized almost to zero even at 150 °C.**

***Index Terms*—Losses, neodymium compounds, permanent magnets, stability.**

## I. Introduction

The number of sintered Nd–Fe–B magnets used in industrial machine applications has increased in recent years. Driving forces towards the permanent magnet technology are the higher achievable power densities and higher efficiencies [1]. This technology requires however a fairly stable magnetic polarization in the permanent magnets through the whole lifetime of the application, typically 20 to 30 years. For this reason, it is important to control the operating environment of magnets.

In rotating machines, induced eddy currents cause a temperature rise, which could be detrimental to the commonly used Nd–Fe–B magnets. Studies of more precise eddy current calculations [2], [3] and of the effects of overheating in fault conditions [4] are attempts towards a better understanding of the operation of these machines. An accurate simulation of the application is required to assist in the selection of the proper magnet material.

Demagnetization effects in Nd–Fe–B magnets occurring in temperature cycling are quite well studied [5]–[7] and documented [8]. In temperature cycling, magnets are at a maximum temperature only for a short time, which represents the overload condition. If the coercive force of the material decreases too much due to the temperature rise and the operating point of the magnet falls below the knee of the demagnetization curve, some permanent flux loss will appear. The term used by the machine designers for the temperature where irreversible losses start to occur is the "critical temperature of the magnet" [9]–[11]. Critical temperature is a characteristic of an application with a certain permeance coefficient (operating point).

Demagnetization curves measured for materials are commonly used in defining the critical temperature of a magnet system. Demagnetization curves, however, consider only the demagnetization occurring immediately. Losses over time are usually assumed to be negligible. The assumption is reasonable when the temperature rise is only temporary. However, continuous eddy currents maintain the operating temperature close to the critical temperature of the magnets, which causes a time-dependent demagnetization through the so-called magnetic viscosity effect.

There are not many published studies on the time-dependent flux losses in commercial sintered Nd–Fe–B magnets. Clegg *et al.* [12] and Mildrum *et al.* [13] have shown that temperature, permeance coefficient (Pc) and coercivity of a magnet affects its time dependent losses. However, in [12] only one temperature and two Pc values were tested for 200 h and in [13], samples were exposed to elevated temperatures in air with a clear effect of oxidation. If proper corrosion protection is applied, the losses with time are only due to the magnetic viscosity.

Magnetic viscosity is known to be a statistical relaxation phenomenon due to thermal fluctuation in the non-equilibrium state of the material [14]. Time dependence of the magnetization $(M)$ as a function of time $(t)$ and magnetic field $(H)$ is described by a logarithmic law [11]

$$M(t,H) = M(t_0,H) - S(H)\ln\frac{t}{t_0} \tag{1}$$

where $S$ is a phenomenological magnetic viscosity constant and $t_0$ is a reference time. Wohlfarth *et al.* [15] have determined the magnetic viscosity constant as

$$S = \frac{kT}{vK} f(H,T) M_S. \tag{2}$$

where $kT$ represents the temperature dependency ($k$ is Bolzmann constant, $T$ is temperature), $vK$ ($v$ is the activation volume, $K$ is the anisotropy constant) depends on the material and its microstructure and $f(H,T)$ is a complicated function, which describes the precise nature of the magnetization process. Thus, magnetization losses over time in permanent magnets depend on the magnetic field, temperature, magnet material and its microstructure and the magnetization history.

In this work, demagnetization losses in commercial sintered Nd–Fe–B magnets were studied. The main interest was in losses

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TABLE I
ROOM TEMPERATURE COERCIVITIES OF THE SAMPLES

| Material | Remanent polarization [T] | Coercivity in room temperature [kA/m] |
|---|---|---|
| 1 | 1.26 | 1240 |
| 2 | 1.19 | 1600 |
| 3 | 1.14 | 1900 |
| 4 | 1.02 | >3000 |

over time and the effects of temperature, coercivity and permeance coefficient on the loss behavior of a magnet. Additionally, a study was made into the effect of a stabilization heat treatment on the losses over time.

## II. METHOD

### A. Temperatures

Samples were kept at five different temperatures. The holding temperatures were room temperature (23 °C), 60 °C, 80 °C, 120 °C, and 150 °C. The elevated temperatures were generated with StabiliTherm EU2 ovens with a temperature accuracy and homogeneity of $\pm 0.5\,^\circ$C.

### B. Material

Four different commercial Nd–Fe–B magnet materials were tested. Intrinsic coercivities at room temperature of these materials are listed in Table I.

### C. Permeance Coefficients

All samples had a rectangular shape with a cross section of 10 mm × 10 mm. The heights of the samples varied according to the tested permeance coefficients $\mathrm{Pc} = 0.33$, 1.1, and 3.3 ($\mathrm{h} =$ 1.6, 4.6, and 10.5 mm, respectively).

During the aging process, the samples were kept in aluminum sample holders, with a distance of 67 mm between the center points of individual samples. The effect of external magnetic fields on the permeance coefficient of each sample, caused by other samples, was evaluated and considered to be negligible.

### D. Measurements

The samples were fully magnetized in a 3.5 T pulsed field. The reference value of the flux of each sample was measured with a Helmholtz coil immediately after the magnetization. To make the plotting of the results easier on a logarithmic time scale, $\mathrm{t}_0$ in (1) was chosen to be 1 h. So, the first loss measurement was carried out after a 1-h stay at the holding temperature. Samples were cooled down to room temperature before the measurements. Flux measurements were carried out at suitable time intervals, at the beginning once a day, and later less frequently, once a week or once a month. Totally, the measurements were carried out at least fifteen times during the 8000-h aging period. The number of identical samples measured at a time varied between 3 and 6.

To minimize oxidation, samples were wrapped in aluminum foil before heating. The foils were removed before the measurements were carried out. After the 8000-h period at 150 °C, the surface area of the samples was checked by optical microscopy and no traces of significant oxidation were observed.

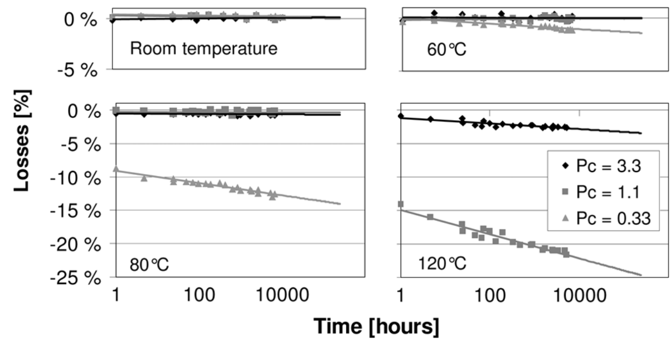


Fig. 1. Flux losses as a function of time at room temperature, 60 °C, 80 °C, and 120 °C for material 1 with room temperature coercivity of 1240 kA/m.

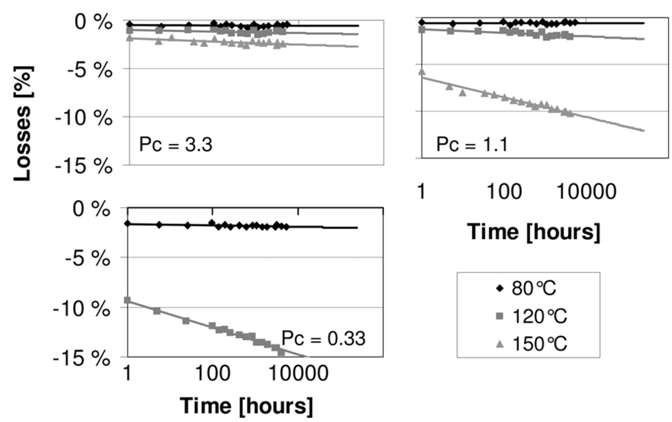


Fig. 2. Flux losses as a function of time for material 2 with room temperature coercivity of 1600 kA/m when $\mathrm{Pc} = 3.3, 1.1,$ and $0.33$.

### E. Stabilization

A stabilization test was performed on a set of samples of material 2. The samples were heat treated at 90 °C for 1 h prior to placing them to the holding temperature of 80 °C. The aim of the stabilization heat treatment is to demagnetize the most unstable domains in the magnet and thus reduce the following losses over time. This means that (1) is not necessarily valid anymore since the initial loss (first term on the right hand side) is achieved at a different temperature than the time dependent losses (second term).

## III. RESULTS

Average measured losses are presented as a function of time in Figs. 1–4. Logarithmic trend curves are plotted in compliance with the measurement points. Trend curves are extended to over 250 000 h, which represents a design life of 30 years.

### A. Effect of Permeance Coefficient

The permeance coefficient had no effect on the losses at room temperature since no losses were detected in any of our sam-

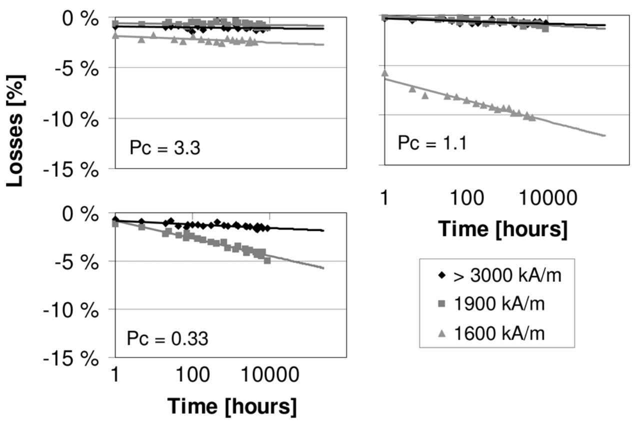


Fig. 3. Flux losses at 150 °C as a function of time for materials 2–4 with different room temperature coercivities when $\mathrm{Pc} = 3.3, 1.1,$ and $0.33$.

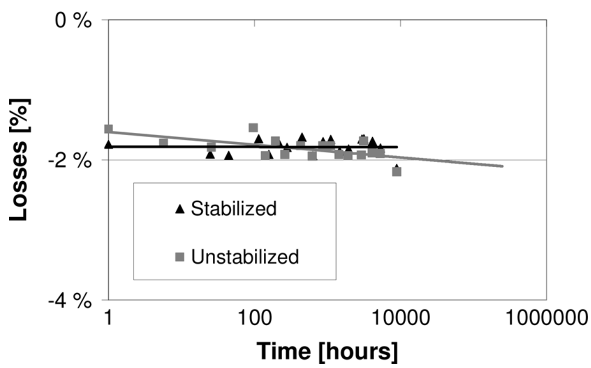


Fig. 4. Loss trends for unstabilized and stabilized samples of material 2 at 80 °C when $\mathrm{Pc} = 0.33$.

ples during the 10 000-h test period. Fig. 1 shows the measured losses for material 1 (see Table I) at room temperature, 60 °C, 80 °C, and 120 °C.

At 60 °C, the trend curve of samples with $\mathrm{Pc} = 0.33$ has slightly bent downwards, but other curves are horizontal. The expected loss after 30 years for $\mathrm{Pc} = 0.33$ samples is less than 2%. At 80°C, the trend curves for samples with $\mathrm{Pc} = 1.1$ and $\mathrm{Pc} = 3.3$ are still horizontal, but the curve for $\mathrm{Pc} = 0.33$ has dropped and reaches as high as 14% in 30 years. At 120 °C, all samples show a downward loss trend. Only in samples with $\mathrm{Pc} = 3.3$ losses in 30 years time are less than 5%. Losses in $\mathrm{Pc} = 1.1$ are estimated to be approximately 25% and $\mathrm{Pc} = 0.33$ samples fall outside the scale.

### B. *Effect of Temperature*

Fig. 2 shows the magnetization losses for material 2 (see Table I) samples with three different Pc values at three different temperatures. When $\mathrm{Pc} = 3.3$, time dependent losses are very small even at 150 °C. No significant differences between samples in different temperatures. The total loss after 30 years is expected to be less than 3%. When $\mathrm{Pc} = 1.1$, clear losses start to occur at 150 °C. However, at 120 °C and 80 °C, losses stay under 3% in 30 years' time. When $\mathrm{Pc} = 0.33$, clear losses occur at as low as 120 °C.

### C. *Effect of Coercivity*

Fig. 3 compares the losses of different materials at 150 °C. When the $\mathrm{Pc} = 3.3$ all three materials are fairly stable. Losses in 30 years are less than 3% for all these materials. When the $\mathrm{Pc} = 1.1$, losses in material 2 increase to about 12% in 30 years. The two other materials show very small losses. When the $\mathrm{Pc} = 0.33$, the total loss after 30 years in material 2 drops to about 32% and it falls outside the scale. The initial loss in the other two materials is minor. In any event, losses over time in material 3, increases the total loss to approximately 6%.

### D. *Effect of Stabilization Heat Treatment*

Fig. 4 compares the loss behavior of stabilized and unstabilized samples at 80 °C. The Pc of the samples was 0.33. The trend curve for losses in stabilized samples is more horizontal, but the difference between these curves is very small. Losses over 2% for both stabilized and unstabilized samples measured after 9000 h of exposure indicates, however, that horizontal loss trend of stabilized samples will change in some point to the downward trend.

## IV. Discussion

The obtained trend curves for flux losses show clearly the division of the irreversible losses into two categories. The intersection point of the trend curve with the loss axis represents the loss occurring immediately at a certain temperature, term $M(t_0)$ in (1). The following trend curve defines the losses over time, $S * \ln(t/t_0)$ in (1). Both terms depend on the coercivity of the material, the permeance coefficient of the magnetic circuit and the temperature. The results also show that when an immediate loss occurs, there will also be losses over time. However, if the immediate loss is negligible, the following losses with time are not necessarily negligible: for example, in Fig. 3 the losses with time in material 3 (at 150 °C when $\mathrm{Pc} = 0.33$) will exceed 5% in 30 years time.

A dramatic difference in the loss behavior of different sized magnets (i.e., different Pc values) can be detected. Near the closed circuit condition $(\mathrm{Pc} = 3.3)$, losses stay within 3% in material 1 $(\mathrm{Hci} = 1240$ kA/m at room temperature) until 120 °C and in other materials until 150 °C. When the Pc is reduced to 1.1, materials with lower coercivities start to show severe losses at temperatures over 120 °C. With a $\mathrm{Pc} = 0.33$, losses occur at as low as 80 °C.

Trout [16] has pointed out that there is a lack of clear definition of the term "maximum operating temperature" for permanent magnets at the moment. Maximum working temperature values given for different materials might be misleading, since the amount of flux loss also depends strongly on the permeance coefficient of the circuit. One expects no losses at the given maximum working temperature. For example, for Chinese magnet grade M, with a minimum coercivity of about 1200 kA/m, the maximum working temperature is defined as 100 °C. However, Fig. 1 shows that already at 80 °C there occur severe losses in such a material if the Pc is small. For the SH grade, with a minimum room temperature coercivity of about 1600 kA/m, the maximum working temperature is defined as 150 °C. However, Fig. 2 shows that at 150 °C, severe losses occur in this kind of material in magnetic circuits with a $\mathrm{Pc} = 1.1$ or less.

However, the room temperature coercivity value is not enough to specify the behavior of a permanent magnet material at elevated temperatures. The coercivity also needs to be defined at the application temperature. The temperature dependence of the coercivity of a permanent magnet material varies for example with grain size [17]. The shape of the $B$–$H$ curve also affects the loss behavior.

Losses over time seem to be reduced with the stabilization heat treatment. The magnetization of the stabilized samples in Fig. 4 can be considered as stable within the first 5000 h. How is the trend curve going to continue after that, is not clear. Fig. 4 also leaves the question of how to define a sufficient stabilization temperature. It is reasonable to expect a stable behavior for the stabilized magnets at shorter exposures than the time at the intersection point of the two trend curves. Thus, the intersection point of these curves should be shifted more to the right to ensure the stability of the magnets for the whole lifetime of the application. This means a higher stabilization temperature and greater initial loss.

## V. Conclusion

Temperature stability of a magnet is demonstrated to be strongly dependent on the coercivity of the permanent magnet material but also on the permeance coefficient of the magnetic circuit.

Flux losses are negligible at low temperatures, where there is no knee in the $B$–$H$ curve or when the operating point is clearly above such knee. When the Pc approaches the knee, time-dependent losses start to occur. Below the knee, the total flux loss increases with a decreasing Pc. However, the proportion of time-dependent losses in the total loss decreases.

According to this study the maximum amount of time-dependent losses expected to occur in commercial sintered Nd–Fe–B magnets in a 30-year period is about 10%. This requires temperatures above the critical temperature and initial losses greater than 10%. However, without exceeding the critical temperature the losses over time can still in some cases be about 6%.

Losses over time during the application could be avoided with proper stabilization heat treatment. The temperature of the heat treatment has to be high enough to ensure the stability of the magnet for the whole life time of the application.

With appropriate design of the circuit and proper selection of the magnet material, the irreversible flux losses in Nd–Fe–B magnets can be minimized down almost to zero. Careful engineering can guarantee the required lifetime for industrial applications in most cases.

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